Shailesh Chaudhari

Software Engineer - TypeScript/Node, backend-strongest, applied AI

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SUMMARY

TypeScript/Node full-stack engineer with 2.5+ years shipping production systems end to end: multi-tenant test-execution infrastructure, idempotent webhook pipelines, real-time platforms, and LLM tool-calling agents inside products. Frontend in React/Next.js, audited to WCAG 2.1 AA. Focused on correctness: concurrency control, deterministic testing, and verification that asserts real outcomes.

EXPERIENCE

Software Engineer, ContextQA — Aug 2025 - Present

- Own the Node.js/TypeScript backend of the core QA-automation platform, a polyglot system: a test-execution engine orchestrating concurrent browser and mobile runs (Playwright, WebdriverIO/Appium, LambdaTest), live browser-session streaming (noVNC) over WebSockets through a Node control plane, and a debug engine with breakpoints and resume-from-checkpoint.
- Primary author of the integrations engine (Node/TypeScript): GitHub App, GitLab and Linear OAuth, Slack Block Kit bot, webhook ingestion. Made it multi-tenant: OAuth handshake state moved from per-pod memory to Redis with an atomic Lua read-and-delete, tenant resolution by installation id, token vault ciphertext-only at rest.
- Node session control plane on GKE: pre-warmed pod pool over Redis, idempotent stop with first-terminal-wins status ordering, request trace ids via AsyncLocalStorage, stable error codes and a diagnostics bundle for on-prem customers, pod eviction protection and memory budgeting.
- Presigned-URL chokepoint so private S3/MinIO/GCS buckets work on-prem; PR-impact analysis flow where a webhook and an MCP tool call share one path.
- Shipped 2 products end to end (React/Next.js + Node.js): Vibe Testing, an AI-assisted UI-testing extension streaming live logs over WebSockets, and AxeTos, a WCAG A/AA/AAA auditor with precise DOM locators. Both still in use.

Software Developer, eSparkBiz Technologies — Aug 2024 - Jul 2025

- Delivered 3 client products full-stack in React and Node.js (EdTech scheduling, sign-language e-learning, corporate LMS), all on schedule; integrated a third-party scheduling API with a Python/CPLEX optimizer, built the real-time calendar UI, and AWS S3 file workflows.

Software Developer Intern, eSparkBiz Technologies — Jan 2024 - Aug 2024

- Built REST APIs and React/Next.js interfaces with authentication and role-based access control (RBAC); integrated Stripe payments and Supabase authentication.

PROJECTS

KhataGO - WhatsApp-first bookkeeping for Indian MSMEs khatago.vercel.app

- Idempotency-guarded reconciliation pipeline: Meta webhook -> DB-enforced dedup -> LLM function-calling (10 tools) -> ledger writes -> Tally XML export. Exactly-once effects over an at-least-once channel via a unique constraint, not a distributed lock: 8 simultaneous deliveries of one message collapse to one stored row; 8 concurrent executors of one agent write step commit exactly one ledger row.
- Containerized (Docker) polling worker over a PostgreSQL-backed queue with an atomic claim, exponential backoff with jitter, and a dead-letter queue (DLQ) with audited replay.
- Next.js/React app in three languages (English, Hindi, Gujarati) with locale-aware invoices and GST reporting; bring-your-own Gemini key encrypted at rest (AES-256-GCM).

BALLAST - deterministic simulation of a multi-tenant session control plane github.com/Shailesh93602/ballast

- Per-tenant concurrency caps, quota windows, and leases over an unreliable substrate; every run is a pure function of one integer seed and replays byte-identically, verified across 1,000 seeds.
- Verified 4 independent ways: 8 invariants, a reference oracle, mutation testing (352 of 362 mutants killed, 291 tests), and a self-verifying shrinker. Found 29 real bugs, 18 of them in the checker rather than the implementation.
- Measured 3 concurrency-control strategies for a flash sale against a real PostgreSQL under 200 concurrent buyers: read-then-write sold 200 of 5 units; conditional update and optimistic versioning each sold exactly 5.

EduScale - real-time coding battles over Socket.io and Redis eduscale.vercel.app | github.com/Shailesh93602/DevScale

- Designed for horizontal scaling across Node.js instances: Socket.io Redis adapter for cross-instance pub/sub, Redlock distributed mutex on battle start, per-key circuit breakers around LLM calls, and Prometheus metrics.
- Next.js/React frontend with role-gated routes, audited to WCAG 2.1 AA by automated axe checks that run on every public route in CI.

SKILLS

- **Languages:** TypeScript, JavaScript, SQL, C++ (DSA)
- **Frontend:** React, Next.js, Tailwind CSS, shadcn/ui, accessibility (WCAG 2.1 AA)
- **Backend:** Node.js, Express, NestJS, Socket.io, Prisma, REST APIs
- **Data:** PostgreSQL, Redis, MongoDB
- **Infrastructure:** Docker / Docker Compose, GitHub Actions (CI/CD), GKE (exposure), Vercel, Supabase, AWS S3
- **AI:** LLM function/tool calling (Gemini), multi-model fallback, per-key isolation, bring-your-own-key encryption, evals, MCP tools, pgvector RAG

EDUCATION

- B.E. in Information Technology, Government Engineering College, Bhavnagar, Gujarat - 2020 to 2024 - CGPA 7.99/10

ACHIEVEMENTS

- 640+ problems solved on GeeksforGeeks
- 5-star C++ on HackerRank
- Finalist, New India Vibrant Hackathon 2023